

Problem 2

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Problem 1 For the following function g defined by

$$g(t) = \begin{cases} 5t + 7 & \text{if } t < -3 \\ \frac{(t-1)(t+2)}{t+2} & \text{if } -3 \leq t < 1 \text{ and } t \neq -2 \\ 4 \ln t & \text{if } t \geq 1 \end{cases}$$

find the **Intervals of Continuity**.

(**Important Note:** Write your answer as a list of intervals, with each interval separated by a comma.)

Explanation. Recall that *Intervals of Continuity* means the largest such intervals on which the function is continuous.

The function g is continuous on $(\infty, -3)$ since, in this interval, $g(t) = 5t + 7$ is a polynomial and therefore continuous on its domain.

g is not continuous at $t = -3$:

$$\lim_{t \rightarrow -3^-} g(t) = \lim_{t \rightarrow -3^-} (5t + 7) = 5(-3) + 7 = -8$$

and

$$\lim_{t \rightarrow -3^+} g(t) = g(-3) = \frac{(-3-1)(-3+2)}{-3+2} = \frac{4}{-1} = -4.$$

Although we have found g is not continuous at $t = -3$, since $\lim_{t \rightarrow -3^+} g(t) = g(-3)$, g is continuous from the right at $t = -3$.

The function g is continuous on $[-3, -2), (-2, 1)$ since, in this interval, $g(t) = \frac{(t-1)(t+2)}{t+2}$ is a rational function whose denominator is not zero. g is not continuous at $t = -2$ because it is not defined there. g is continuous on $(1, \infty)$, since, on this interval, $g(t) = 4 \ln(t)$ is a constant multiple of a logarithmic function.

g is continuous at $t = 1$:

$$\lim_{t \rightarrow 1^-} g(t) = \lim_{t \rightarrow 1^-} \frac{(t-1)(t+2)}{t+2} = \frac{0}{3} = 0,$$

and

$$\lim_{t \rightarrow 1^+} g(t) = g(1) = 4 \ln(1) = 0$$

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So, the Intervals of Continuity are

$$(-\infty, -3), [-3, -2), (-2, \infty)$$

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